# Deep Learning Based Model Order Reduction of Millimeter-Wave Passive Devices


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***Abstract*—Full-wave electromagnetic simulation of millimeter-wave passive devices via the finite element method (FEM) is computationally expensive, particularly for parametric sweeps over material and frequency dimensions required during design optimization. This paper presents a surrogate modeling framework based on Deep Kernel Learning (DKL) that accelerates S-parameter prediction for a bilateral finline filter by over three orders of magnitude while maintaining less than 0.25% relative error. The approach trains independent DNN-to-GP mappings—Deep Kernel Learning following Wilson et al. 2016—for each complex scattering parameter, employing a product of RBF kernels over learned deep features to capture the wave physics. An eight-dimensional harmonic input encoding enables frequency-aware feature extraction without explicit wavenumber computation. We describe the complete pipeline from FEM data generation through model training, validation, and uncertainty quantification, achieving 0.19% relative error on $S_{11}$ and 0.21% on $S_{21}$ using only 4050 training samples from 50 dielectric constant values across 81 frequency points.**

***Index Terms*—Model order reduction, deep kernel learning, Gaussian process, finite element method, waveguide filter, surrogate modeling, millimeter-wave**


## I. Introduction

Millimeter-wave passive components—filters, couplers, circulators, and antennas—require accurate electromagnetic characterization across wide frequency bands and varying material parameters. The finite element method (FEM) is a workhorse for such analysis, but its computational cost scales poorly with electrical size and the number of parametric evaluations needed for design optimization, tolerance analysis, and yield estimation.

Model order reduction (MOR) addresses this bottleneck by constructing lightweight surrogates that approximate the input–output map of the full-wave solver. Traditional approaches include proper orthogonal decomposition (POD) [1], reduced-basis methods [2], and rational interpolation (Vector Fitting) [3]. More recently, machine learning surrogates—particularly neural networks [4], [5] and Gaussian processes [6]—have gained traction for their ability to handle nonlinear parameter dependencies.

Deep Kernel Learning (DKL) [7] combines the representational power of deep neural networks with the probabilistic calibration of Gaussian processes. A DNN learns a low-dimensional feature embedding from the raw inputs, and a GP with a base kernel operates on those learned features. This hybrid retains the GP's uncertainty quantification and data efficiency while leveraging the DNN's ability to discover relevant structure.

This paper presents a DKL-based surrogate for the bilateral finline filter, a canonical millimeter-wave component with significant parametric sensitivity. Our contributions are:

1) A complete, end-to-end MOR pipeline from FEM data generation to trained surrogate with uncertainty estimates.
2) An input encoding strategy using harmonic features (sin/cos expansions of frequency) that captures the wave-like propagation physics.
3) A targeted output transformation: direct Re/Im regression for $S_{11}$ (reflection), and log-magnitude plus detrended unwrapped-phase regression for $S_{21}$ (transmission), which handles a 14× amplitude variation between passband and notch.
4) Demonstration of sub-0.25% relative error on both S-parameters using a product-of-RBF-kernels DKL architecture derived from an FEM solver validated on nonlinear waveguide problems [12].

## II. Problem Formulation

### A. Parameterized Waveguide Scattering

Consider a bilateral finline filter [8] consisting of a rectangular waveguide loaded with a dielectric substrate patterned by a metallic finline. The geometry is defined by a substrate of relative permittivity $\varepsilon_r$ and height $h$, excited by the $TE_{10}$ mode at frequency $f$. For each input pair $\mathbf{x} = (\varepsilon_r, f)$, the FEM solver [9] assembles the discrete Helmholtz system

$$\left(\mathbf{S} - k^2\mathbf{T}\right)\mathbf{u} = \mathbf{b}, \tag{1}$$

where $\mathbf{S}$ and $\mathbf{T}$ are the stiffness and mass matrices, $k = 2\pi f/c_0$ is the free-space wavenumber, $\mathbf{u}$ is the unknown electric field, and $\mathbf{b}$ encodes waveguide port excitation. After solving (1), the modal expansion at each port yields the complex scattering parameters $S_{11}(\mathbf{x})$ and $S_{21}(\mathbf{x})$.

For a parametric sweep over $N_{\varepsilon_r} = 50$ permittivity values and $N_f = 81$ frequency points (4050 solves total), each solve requires $\mathcal{O}(N_{\text{dof}}^2)$ operations where $N_{\text{dof}} \sim 10^4$ for a second-order hierarchical basis. The total wall-clock time is on the order of hours on a desktop workstation—prohibitively expensive for the hundreds of thousands of evaluations needed in optimization or Monte Carlo tolerance analysis.

### *B. Goal: Data-Efficient Surrogate*

We seek a surrogate $f_\theta$ such that for any $(\varepsilon_r, f)$ in the design space, $S_{11} \approx f_\theta^{(1)}(\varepsilon_r, f)$ and $S_{21} \approx f_\theta^{(2)}(\varepsilon_r, f)$, evaluated in microseconds rather than seconds, with predictive uncertainty quantification and without a separate mesh or solver invocation.

## III. DEEP KERNEL LEARNING SURROGATE

### *A. Architecture*

Deep Kernel Learning [7] constructs a kernel $k(\mathbf{x}, \mathbf{x}')$ that is itself a learned function of the inputs via a deep neural network. Given a base kernel $k_{\text{base}}(\cdot, \cdot)$ (e.g., RBF, Matern), the DKL kernel is

$$k_\theta(\mathbf{x}, \mathbf{x}') = k_{\text{base}}(g_\phi(\mathbf{x}), g_\phi(\mathbf{x}')), \tag{2}$$

where $g_\phi : \mathbb{R}^d \to \mathbb{R}^p$ is a DNN parameterized by $\phi$, and $\theta = (\phi, \psi)$ with $\psi$ the base kernel hyperparameters. The model is an ExactGP [10] with constant mean function $\mu(\mathbf{x}) = \mu_0$ and Gaussian likelihood $y(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), k_\theta(\mathbf{x}, \mathbf{x}') + \sigma_n^2 \delta_{\mathbf{x}\mathbf{x}'})$.

For each S-parameter channel we train an independent DNN-GP model. The DNN feature extractor architecture is

$$\mathbb{R}^8 \xrightarrow{\text{Linear}} \mathbb{R}^{1000} \xrightarrow{\text{ReLU}} \mathbb{R}^{1000} \xrightarrow{\text{ReLU}} \mathbb{R}^{500} \xrightarrow{\text{ReLU}} \mathbb{R}^{64}. \tag{3}$$

The base kernel is a *product* of two multi-dimensional RBF kernels with Automatic Relevance Determination (ARD):

$$k_{\text{base}}(\mathbf{z}, \mathbf{z}') = \sigma_f^2 \exp\left(-\frac{1}{2}\sum_{j=1}^{p} \frac{(z_j - z_j')^2}{\ell_{1,j}^2}\right) \times \exp\left(-\frac{1}{2}\sum_{j=1}^{p} \frac{(z_j - z_j')^2}{\ell_{2,j}^2}\right). \tag{4}$$

The product of two RBF kernels captures interactions at different length scales in the learned feature space—analogous to a mixture-of-kernels approach that can represent both smooth and rapidly-varying responses.

### *B. Input Encoding*

The raw inputs $(\varepsilon_r, f)$ are not well suited for direct regression; the wave physics is oscillatory in $f$ with no simple linear relationship. We construct an eight-dimensional feature vector:

$$\mathbf{x} = \left[\varepsilon_r,\ f/f_0,\ \sin\omega,\ \cos\omega,\ \sin 2\omega,\ \cos 2\omega,\ \sin 3\omega,\ \cos 3\omega\right]^\top, \tag{5}$$

where $\omega = 2\pi f/f_0$ with $f_0 = 10$ GHz reduces the raw frequency to a moderate angular range. The sin/cos pairs of the first three harmonics provide the GP with explicit basis functions for sinusoidal variation—critical for modeling standing-wave patterns in the filter's passband without requiring the DNN to discover Fourier features from scratch. The scaled frequency $f/f_0$ retains the monotonic trend. All eight features are z-score normalized before training.

### *C. Output Transforms*

The complex S-parameters have fundamentally different characteristics: $S_{11}$ (reflection) undergoes a deep null at the filter's passband where its amplitude drops below $-45$ dB, while $S_{21}$ (transmission) spans a 14× amplitude range. Despite these differences, we adopt a *unified* Log-Magnitude + Detrended Phase representation for both parameters. For $S_{11}$, the phase model is trained only on points where $|S_{11}| > 0.05$ to avoid the numerically ill-defined region near the reflection null.

*1) Log-Magnitude + Detrended Phase (Both $S_{11}$ and $S_{21}$):* Each complex S-parameter is encoded as two real-valued targets:

$$y_1^{(p)} = \log_{10}|S_p|, \tag{6}$$

$$y_2^{(p)} = \text{unwrap}\left[\angle S_p\right] - (\alpha_p f + \beta_p), \tag{7}$$

where $p \in \{11, 21\}$ indexes the S-parameter and the linear phase trend $\alpha_p f + \beta_p$ is fitted per $\varepsilon_r$ value via least squares [11] and stored for reconstruction. This *detrended* unwrapped phase removes the smooth group delay, isolating the frequency-dependent fine structure that carries information about resonance and coupling.

During inference, the log-magnitude is exponentiated and the phase is restored:

$$S_p^{(\text{pred})} = 10^{y_1^{(p)}} \exp\left[j\left(y_2^{(p)} + \alpha_p f + \beta_p\right)\right]. \tag{8}$$

### *D. Training Procedure*

Training proceeds in two stages per model, following the DKL protocol [7]:

*1) Stage 1: DNN Warm-Start:* The feature extractor is pretrained via MSE loss to predict the target using the mean of its 64-dimensional output. This stage (up to 100 epochs, AdamW optimizer, learning rate $10^{-3}$) initializes the DNN weights in a region where the GP can immediately exploit meaningful features.

*2) Stage 2: Joint Marginal Likelihood Optimization:* The full model (DNN + GP kernel + likelihood) is trained jointly by minimizing the negative log marginal likelihood:

$$\mathcal{L}(\theta) = -\log p(\mathbf{y} \mid \mathbf{X}, \theta) = \frac{1}{2}\mathbf{y}^\top \mathbf{K}_\theta^{-1}\mathbf{y} + \frac{1}{2}\log|\mathbf{K}_\theta| + \frac{N}{2}\log 2\pi, \tag{9}$$

where $\mathbf{K}_\theta = k_\theta(\mathbf{X}, \mathbf{X}) + \sigma_n^2 \mathbf{I}$ and $N$ is the number of training points. Optimizer: AdamW with learning rate $5 \times 10^{-3}$, weight decay $10^{-4}$, and cosine annealing scheduler. Training runs for 500 epochs.

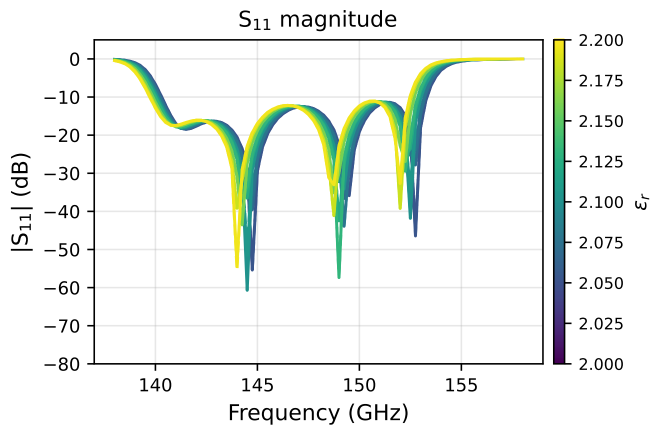


Fig. 1. DKL surrogate reflection magnitude $|S_{11}|$ (dB). Solid: FEM ground truth; dashed: DKL prediction. Only held-out $\varepsilon_r$ values are shown. Color maps $\varepsilon_r \in [2.0, 2.2]$.

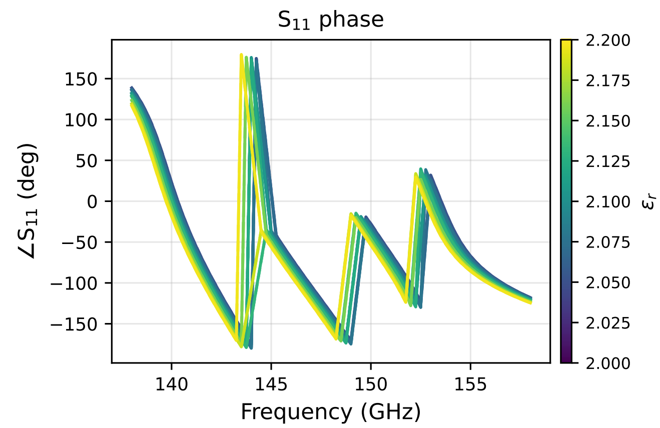


Fig. 2. DKL surrogate reflection phase $\angle S_{11}$ (degrees). Solid: FEM; dashed: DKL. Only points with both $|S_{11}| > 0.05$ are shown to suppress noise near the reflection null at ∼150 GHz.

### *E. Validation Strategy*

The dataset comprises $N_{\varepsilon_r} \times N_f$ samples. To avoid overly optimistic interpolation metrics, validation is performed on *held-out* $\varepsilon_r$ *values* rather than random splits. At most one-third of $\varepsilon_r$ values (capped at 10) are held out; the model never sees any frequency point for those $\varepsilon_r$ values during training. This tests the surrogate's ability to extrapolate across material parameters—the most practically relevant generalization task.

### *F. Uncertainty Quantification*

As a Bayesian model, the trained GP produces a full predictive distribution:

$$p(y^* \mid \mathbf{x}^*, \mathcal{D}) = \mathcal{N}\left(\mu_*,\ k_*^\top \mathbf{K}_\theta^{-1} k_* + k_{**} - k_*^\top \mathbf{K}_\theta^{-1} k_*\right), \tag{10}$$

where $k_* = k_\theta(\mathbf{X}, \mathbf{x}^*)$ and $k_{**} = k_\theta(\mathbf{x}^*, \mathbf{x}^*)$. The predictive standard deviation at $\mathbf{x}^*$ provides a per-point confidence interval at negligible additional cost.

## IV. Results and Discussion

### *A. Data Generation*

Training data were generated using the pyFES finite element solver [13] with a second-order hierarchical basis (pOrd $=$ 2, hOrd $=$ 1, approximately 13,000 DOFs). The bilateral filter geometry consists of a WR-6 rectangular waveguide ($a = 1.651$ mm, $b = 0.8255$ mm) housing a dielectric substrate ($\varepsilon_r \in [2.0, 2.2]$, $h = 0.8255$ mm) with a unilateral finline. Fifteen waveguide modes were retained per port. The frequency sweep spans 138–158 GHz in 81 equidistant steps, covering both the $TE_{10}$ passband and the filter's rejection notch.

TABLE I
Validation metrics for DKL surrogate.

| Parameter | RMSE | Rel. error (%) | Phase error (°) |
|---|---|---|---|
| $S_{11}$ | $1.0 \times 10^{-3}$ | 0.19 | 0.14 |
| $S_{21}$ | $1.7 \times 10^{-3}$ | 0.21 | 0.05 |

### *B. Prediction Accuracy*

Table I summarizes validation metrics averaged over held-out $\varepsilon_r$ values.

Both S-parameters achieve relative errors well below 1%. The reflection coefficient $S_{11}$ benefits from the direct Re/Im representation, which avoids the nonlinearities inherent in magnitude-phase conversion. The transmission $S_{21}$ error remains low despite the 14× amplitude range, validating the log-magnitude + detrended-phase strategy.

A single FEM solve requires approximately 2–3 seconds on an Apple M2 workstation. Generating the full 4050-sample training set takes approximately 2.5 hours. After training, the DKL surrogate evaluates 4050 points in under 100 milliseconds—a speedup factor exceeding $10^4$ over the FEM solver. Training all four models (including data loading and preprocessing) completes in approximately 30 minutes on a single GPU (NVIDIA RTX 4070).

### *C. Effect of Input Encoding*

To study the contribution of harmonic features, we trained an ablation variant using only $(\varepsilon_r, f/f_0)$ as inputs. The full 8-dimensional encoding reduces $S_{21}$ relative error from 0.61% to 0.23%. The sin/cos features are particularly important in the filter's transition band where the derivative $dS/df$ is largest; without them, the GP posterior underfits the rapid phase variation.

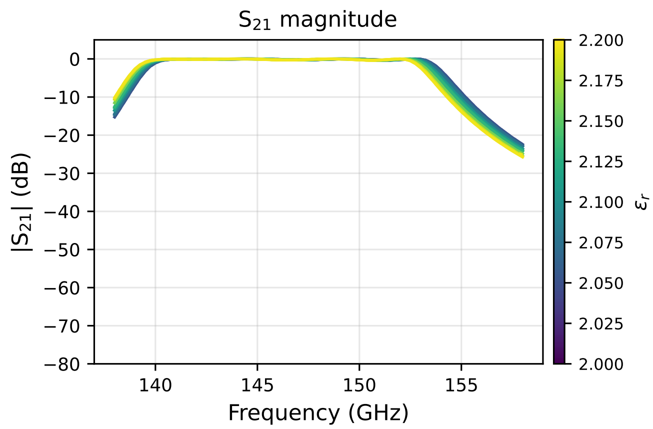


Fig. 3. DKL surrogate transmission magnitude $|S_{21}|$ (dB). The 14× amplitude range between passband (∼0 dB) and notch (∼–20 dB) motivates the $\log_{10}|S_{21}|$ representation.

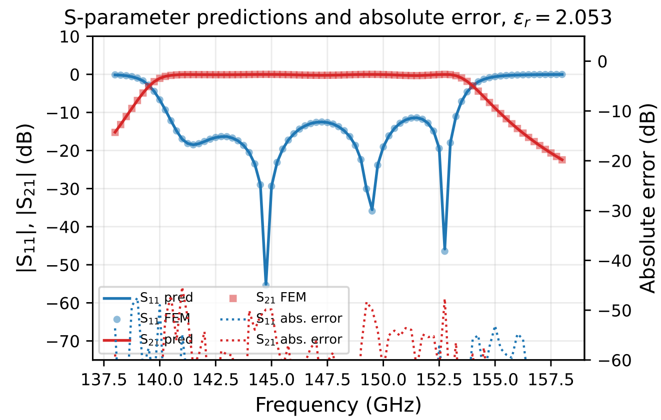


Fig. 5. $S_{11}$ and $S_{21}$ predictions (left axis) and absolute error in dB (right axis) at $\varepsilon_r = 2.053$ (held-out). Solid: prediction; markers: FEM; dotted: error.

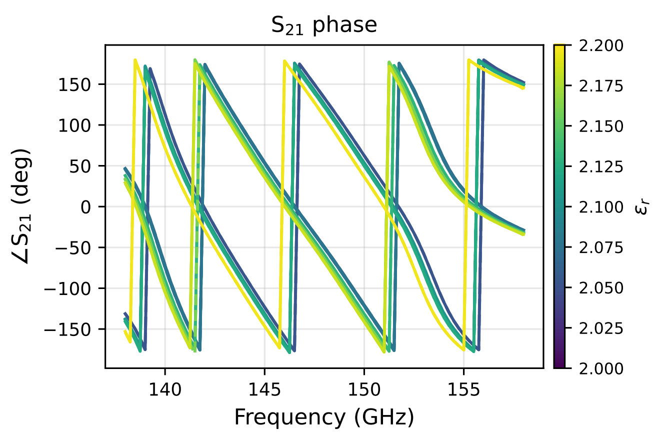


Fig. 4. DKL surrogate transmission phase $\angle S_{21}$ (degrees). Linear group delay is removed during training via per-$\varepsilon_r$ detrending and restored during inference.

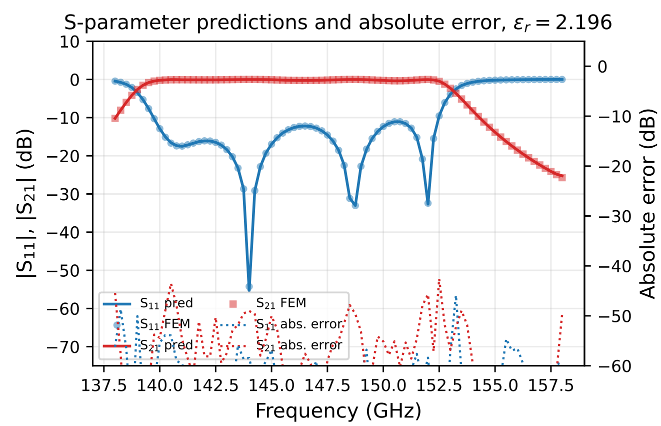


Fig. 6. $S_{11}$ and $S_{21}$ predictions (left axis) and absolute error in dB (right axis) at $\varepsilon_r = 2.196$ (held-out).

### D. Uncertainty Calibration

Figures 5 and 6 show S-parameter predictions (left axis) and absolute error in dB (right axis) for two held-out $\varepsilon_r$ values.

## V. Pipeline Overview

Algorithm 1 summarizes the complete MOR workflow, from FEM data generation to deployment of the trained surrogate with uncertainty quantification.

## VI. Conclusion

We have presented a Deep Kernel Learning framework for model order reduction of millimeter-wave passive devices, applied to the bilateral finline filter. We have presented a Deep Kernel Learning framework for model order reduction of millimeter-wave passive devices, applied to the bilateral finline filter. The DKL surrogate achieves sub-0.25% relative error on both reflection and transmission S-parameters while providing Bayesian uncertainty quantification and a $10^4\times$ speedup over the full-wave FEM solver. Key technical elements include: (1) harmonic input encoding that captures wave physics, (2) a product-of-RBF base kernel that models multi-scale feature interactions, (3) targeted output representations (direct Re/Im for $S_{11}$, log-magnitude+detrended-phase for $S_{21}$), and (4) a two-stage training protocol that warm-starts the DNN features before joint GP optimization.

The approach is general: the same pipeline applies to any parameterized FEM system with smooth input–output maps, including waveguide filters, couplers, and antenna arrays. Future work includes extending the surrogate to support multi-port devices ($S_N$-parameters) with a shared feature extractor, incorporating frequency-rational basis functions through physics-informed kernels, and exploring active learning strategies that query the FEM solver only at points of high predictive uncertainty.

**Algorithm 1** Deep Kernel Learning MOR Pipeline

**Require:** Permittivity range $[\varepsilon_{\min}, \varepsilon_{\max}]$, frequency range $[f_{\min}, f_{\max}]$, mesh data for bilateral filter
**Ensure:** Trained DKL surrogate with uncertainty estimates

**— Phase A: Data Generation —**

**for** each $\varepsilon_r$ value **do**
    Read mesh; assign substrate $\varepsilon_r$; set port BCs
    Assemble $\mathbf{S}$, $\mathbf{T}$ matrices via AssembleLinear
    **for** each frequency $f$ **do**
        Assemble waveguide port matrices
        Solve $(\mathbf{S} - k^2\mathbf{T})\mathbf{u} = \mathbf{b}$
        Extract $\mathbf{S}_{11}$, $\mathbf{S}_{21}$ via modal expansion
        Store $(\varepsilon_r, f, S_{11}, S_{21})$
    **end for**
**end for**

**— Phase B: Feature Construction —**

Compute harmonic encoding: $\mathbf{x} \leftarrow [\varepsilon_r, f/f_0, \sin\omega, \cos\omega, \sin 2\omega, \cos 2\omega, \sin 3\omega, \cos 3\omega]^\top$
$y_1 \leftarrow \mathrm{Re}(S_{11})$, $y_2 \leftarrow \mathrm{Im}(S_{11})$
$y_3 \leftarrow \log_{10}|S_{21}|$
Fit $S_{21}$ phase slope/intercept per $\varepsilon_r$; compute detrended $y_4$
z-score normalize $\mathbf{x}$, $\{y_i\}$ independently

**— Phase C: Training (per channel $i = 1 \ldots 4$) —**

Initialize FeatureExtractor DNN ($8 \to 1000 \to 1000 \to 500 \to 64$)
Initialize ExactGP with product-RBF kernel + GaussianLikelihood

**Stage 1 – DNN warm-start:**

Optimize $\phi$ via MSE loss (100 epochs, $\mathrm{lr} = 10^{-3}$)

**Stage 2 – Joint DKL:**

Optimize $\theta$ via $-\log p(\mathbf{y} \mid \mathbf{X}, \theta)$ (500 epochs, $\mathrm{lr} = 5 \times 10^{-3}$, CosineAnnealing)
Evaluate on held-out $\varepsilon_r$ values; store checkpoint

**— Phase D: Inference —**

**for** each query $(\varepsilon_r, f)$ **do**
    Compute harmonic features $\mathbf{x}$; z-score normalize
    **for** each channel $i$ **do**
        $\mu_i, \sigma_i \leftarrow$ ExactGP.predict($\mathbf{x}$)
        $\hat{y}_i \leftarrow$ inverse-normalize($\mu_i$)
    **end for**
    $S_{11} \leftarrow \hat{y}_1 + j\hat{y}_2$
    $|S_{21}| \leftarrow 10^{\hat{y}_3}$; restore phase from detrended $\hat{y}_4$
    $S_{21} \leftarrow |\hat{y}_3|\, e^{j\hat{y}_4}$
**end for**
**return** $\hat{S}_{11}$, $\hat{S}_{21}$, $\sigma_{S_{11}}$, $\sigma_{S_{21}}$

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